

DATABASE SYSTEMS PROJECT

DATE:11-9-2026

TrafficFlowDB: A Database Management System for Real-Time Traffic Monitoring, Congestion Analysis, and Intelligent Route Optimization

J.B.ROHIT PRAGADEESH -25BDS0276

RAGHAVAPRIYAN SARAVANAPRIYAN - 25BCE2513

ADITYA MANOJ KUNTE - 25BAI0123

TEAM ROLES AND RESPONSIBILITIES

Team Member	Role	Responsibilities
Member 1	Technical Lead & Database Developer	• Design and implement PostgreSQL database, ER/EER and normalization • Develop SQL queries, indexes, views, triggers, stored procedures and database integration
Member 2	Backend & Traffic Analytics Developer	• Develop FastAPI backend, GPS simulator and GPS data processing • Implement traffic analysis, congestion detection and Dijkstra-based route optimization
Member 3	Frontend, Dashboard & Testing Developer	• Develop live dashboard, map interface, vehicle visualization and route display • Handle API integration, system testing, documentation and presentation

Abstract

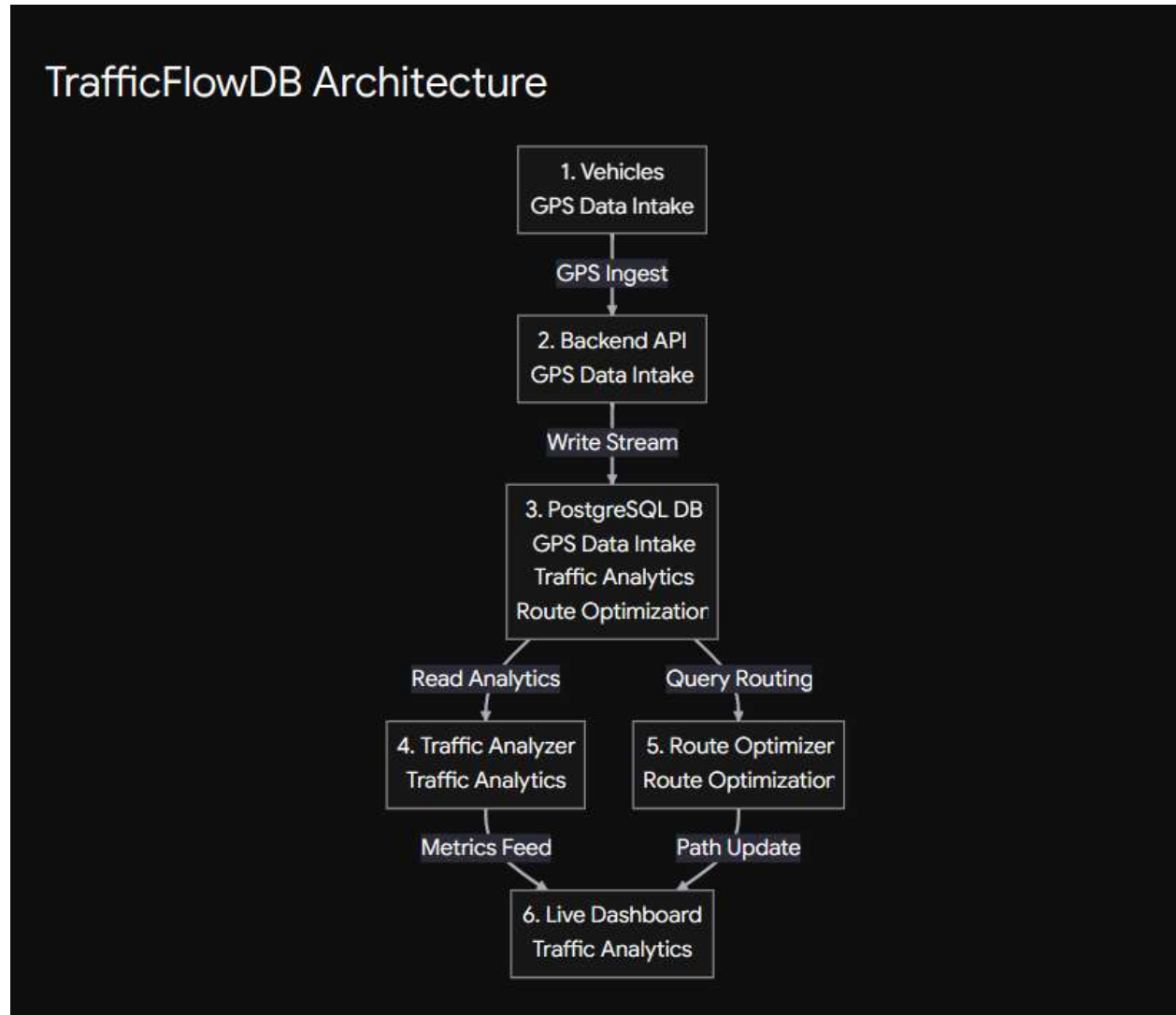
Traffic congestion is a major challenge in modern urban transportation systems, resulting in increased travel time, fuel consumption, and environmental pollution. Efficient management of traffic data requires systems capable of storing, processing, and analyzing large volumes of continuously changing information. TrafficFlowDB is a Database Management System designed for real-time traffic monitoring, congestion analysis, and intelligent route optimization.

The system collects simulated or real-time GPS data from multiple vehicles and stores the information in a centralized relational database. The road network is represented using road segments and intersections, while vehicle location and speed information are continuously recorded as traffic observations. The system analyzes the collected data to determine traffic density, average vehicle speed, and congestion levels on individual road segments.

Based on the current traffic conditions and the distance between locations, TrafficFlowDB calculates the estimated travel cost of different routes. A route optimization module applies graph-based algorithms to identify the most efficient path between a given source and destination. The recommended route is dynamically influenced by both geographical distance and real-time traffic congestion.

The project demonstrates the practical role of Database Management Systems in smart transportation applications. It integrates database design, real-time data processing, traffic analytics, and graph-based route optimization into a unified system. TrafficFlowDB can serve as a foundation for future intelligent transportation systems involving live GPS tracking, predictive traffic analysis, emergency route planning, and smart city infrastructure.

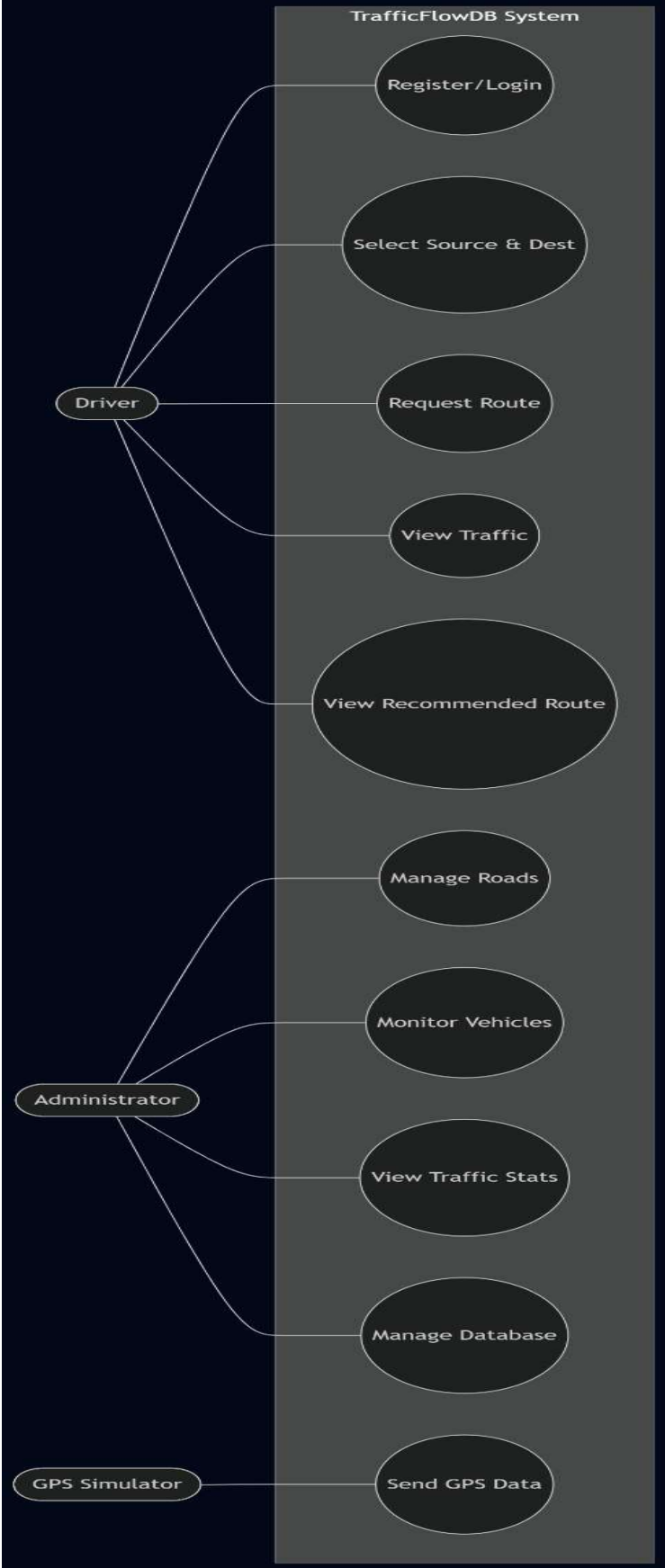
TrafficFlowDB



Uml diagrams:

1. Use Case Diagram

- **What it covers:** This topic defines the scope of the system by illustrating how different entities (Drivers, Administrators, and the automated GPS Simulator) interact with the core features of the database, from managing infrastructure to requesting dynamic routes.



> **Driver** can register/login, select source and destination, request routes, view traffic, and view recommended routes.

> **Administrator** can manage routes, monitor vehicles, view traffic status, and manage the database.

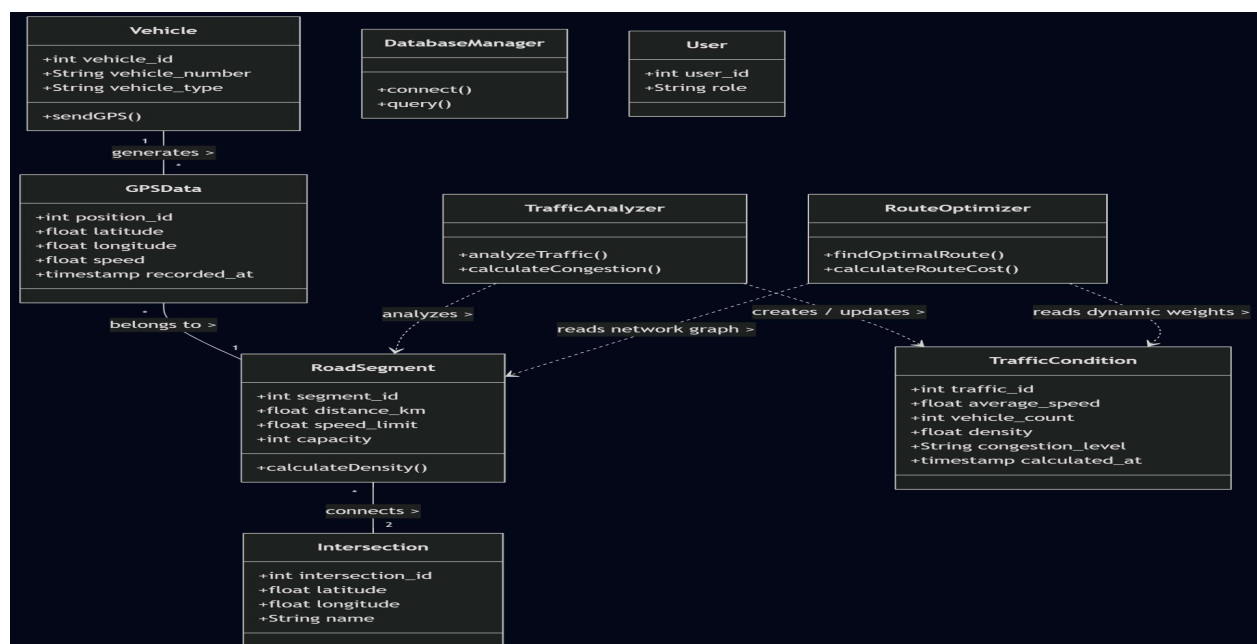
>The diagram shows how different **users interact with the TrafficFlowDB system**.

>It separates the responsibilities of the **Driver and Administrator** clearly.

>Overall, it represents the **functional requirements and user interactions** of the system.

2. Class Diagram

- **What it covers:** This topic outlines the static blueprint of the application. It highlights the primary data entities (Vehicles, Intersections, Road Segments) alongside the processing engines (Traffic Analyzer, Route Optimizer) to show how data is structured and linked before it enters PostgreSQL.



>The Class Diagram represents the **static structure** of the TrafficFlowDB application.

>Major classes include **Vehicle, User, RouteManager, TrafficAnalyzer, RouteOptimizer, GPSTData, TrafficCondition, and Route.**

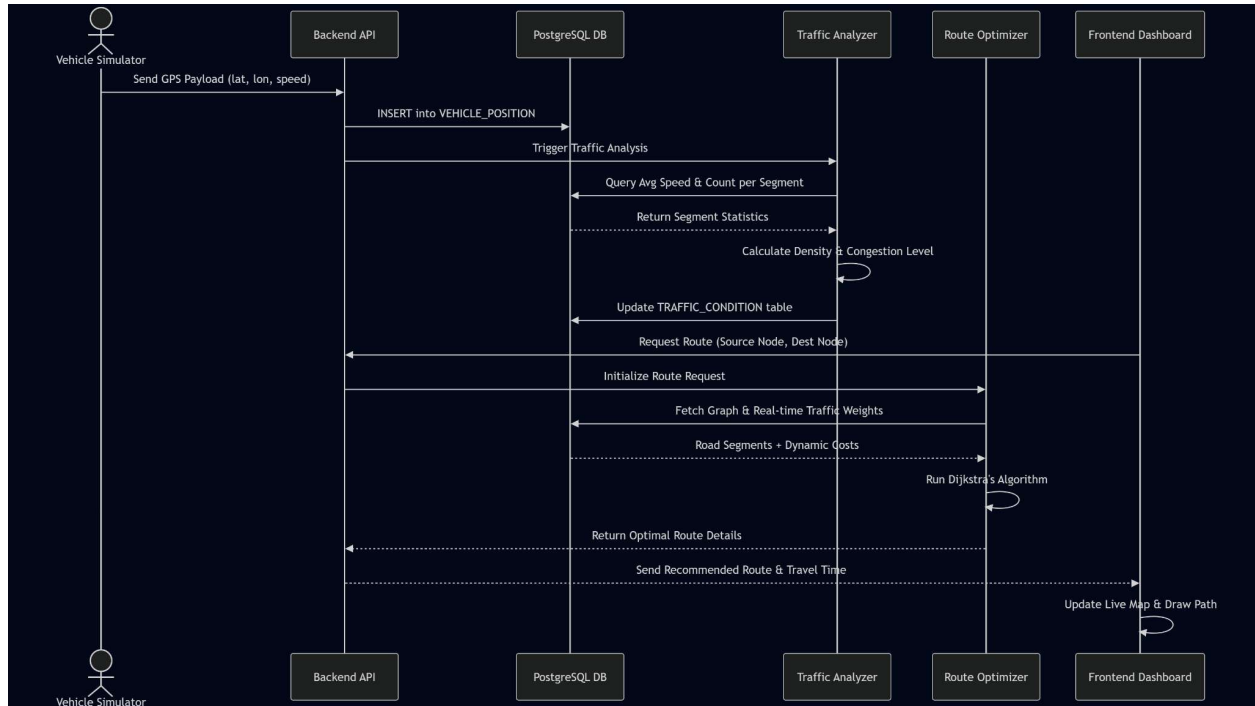
>**Traffic Analyzer** processes traffic information and supports congestion analysis.

>**RouteOptimizer** is responsible for route optimization based on traffic conditions.

>The diagram shows the **relationships between application classes** before the data is stored in PostgreSQL.

3. Sequence Diagram

- **What it covers:** This topic demonstrates the live, temporal aspect of your project. It steps through the exact chronological lifecycle of a single GPS ping—from the vehicle simulator, into the database, through the congestion analytics, and finally triggering a Dijkstra-based route optimization for the end user.



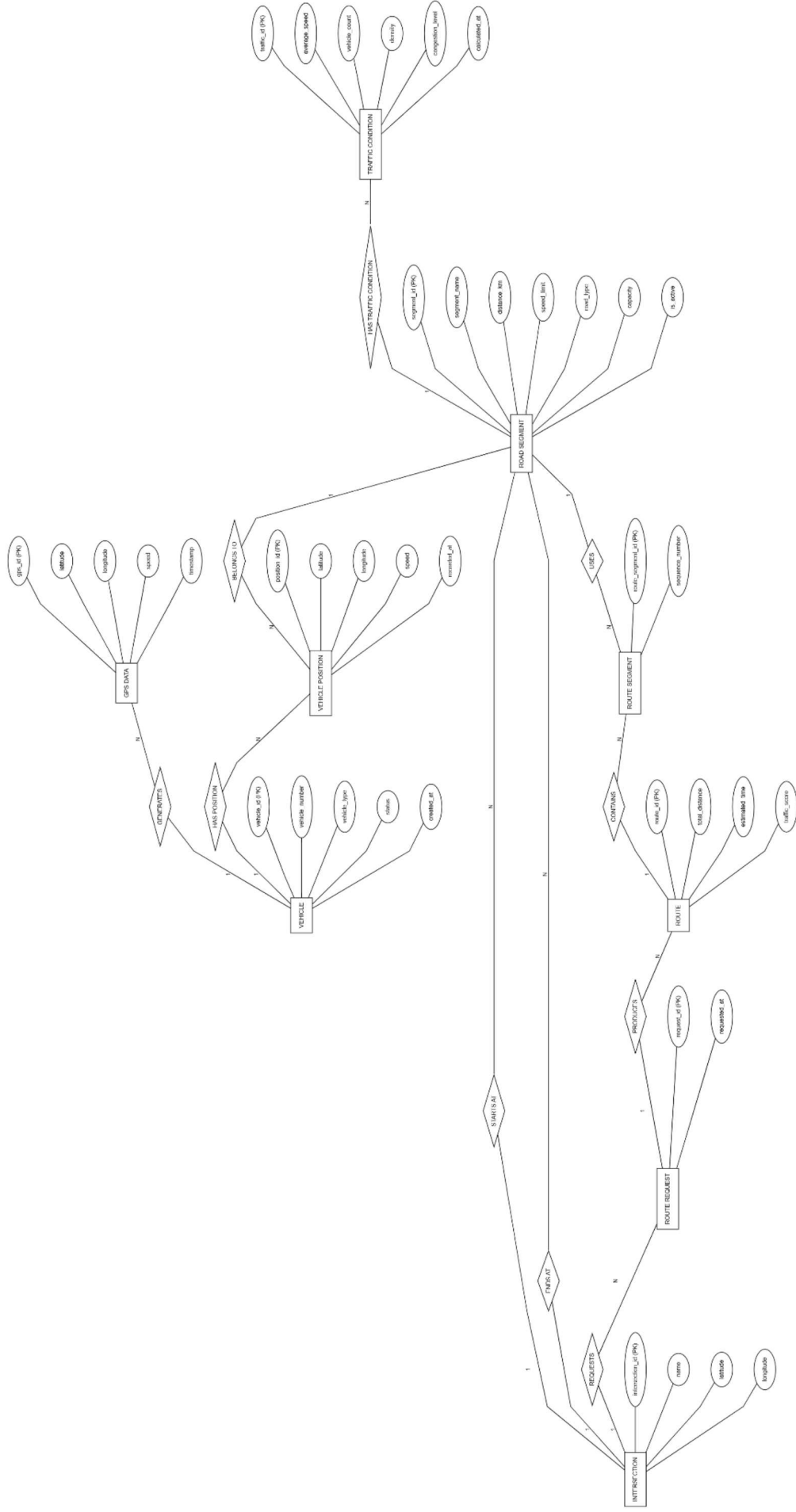
>The Sequence Diagram represents the **time-based flow of operations** in the system.

>The process begins with the **user requesting a route** by providing source and destination.

>Vehicle/GPS information is processed and relevant **traffic data is retrieved from the database**.

>The system performs **congestion analysis and route optimization**. Finally, the optimized/recommended route is **returned and displayed to the user**

ER Diagram



>The ER Diagram represents the **database entities and their relationships**.

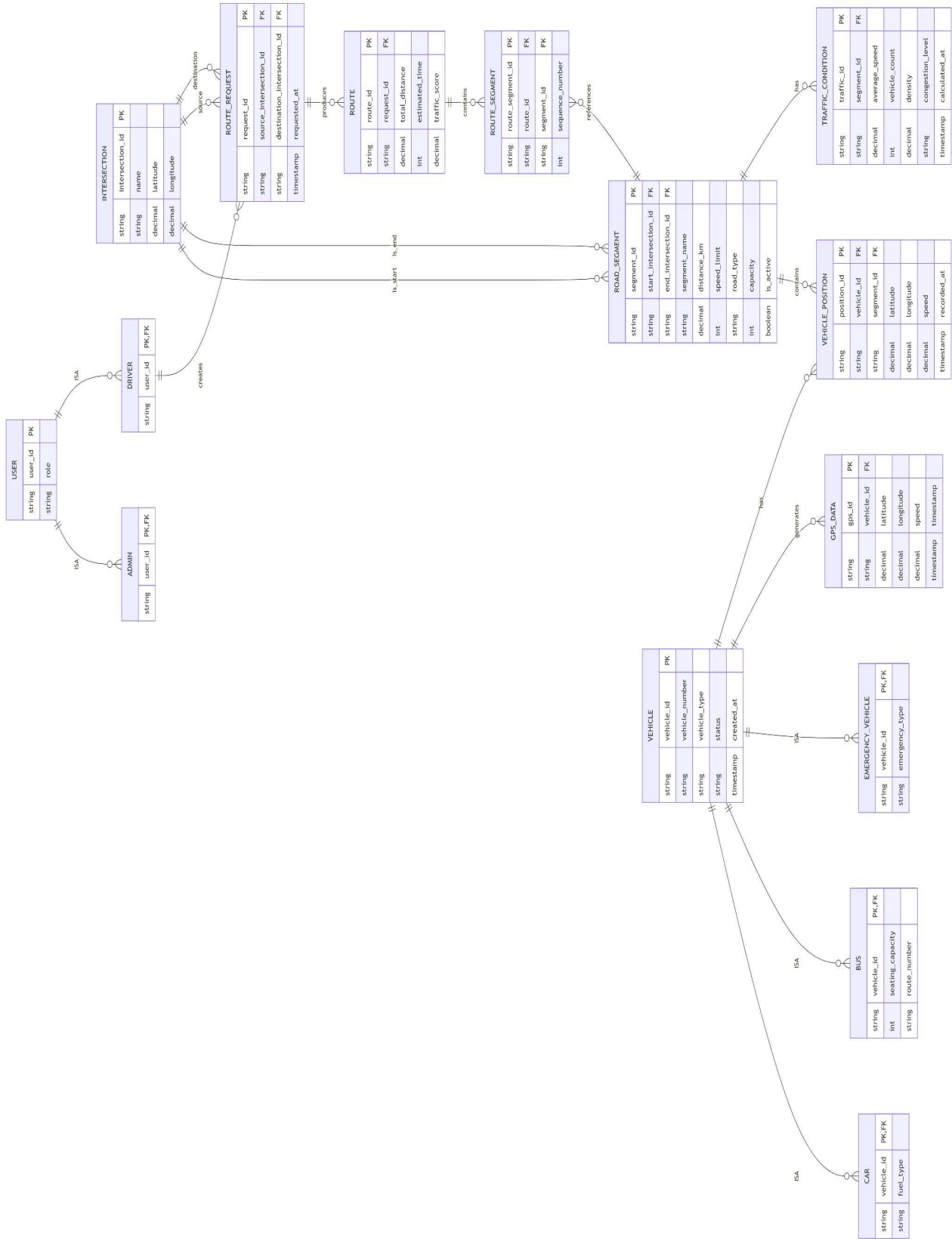
>It organizes the major data components required for **traffic monitoring and route management**.

Entities are connected through relationships to show how different pieces of traffic data are associated.

>The structure supports storing information related to **vehicles, routes, traffic conditions, GPS data, and users**.

> It provides the **conceptual database design** that forms the foundation for implementing the PostgreSQL database.

EER DIAGRAM



>The EER Diagram provides a **more detailed representation of the database structure** than the basic ER model.

>It shows entities with their **attributes, keys, and relationships**.

The diagram represents the connections between different database components used by TrafficFlowDB.

>It helps define how the database should be **organized and implemented efficiently**.

>Overall, it serves as a detailed blueprint for the **PostgreSQL database design and implementation**.
